



A risk-aware assessment for buffer recycling across unit operations for monoclonal antibody purification and its potential

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ABSTRACT

Water for injection (WFI) for biopharmaceuticals is coming into focus considering the sustainability goals of society. WFI production consumes excessive amounts of water and energy. Strategies to reduce WFI energy cost include cold WFI production or reduction of buffer volumes used in the process. A suggested alternative with proposed saving of up to 90 % was the recycling of buffers, potentially across multiple unit operations. In this work a risk-based assessment of the idea is made to quantify the potential savings. All streams of an antibody purification were classified according to their potential risk. This analysis showed that the process does not produce sufficient recyclable outlet streams to cover all replaceable inlet streams. From a potentially replaceable 42 % of inlet buffers only a total of 23 % can be replaced, even if cross-batch recycling is allowed. If this is not allowed, the savings drop down to 14 %. While this saves some water and energy, it shows that making full use of the theoretical potential is impossible, and 90 % reduction is never achievable when risk-aware design of recycling across unit operations is used. A universal risk-based assessment is important and can build the rationale to convince regulatory authorities in case of implementation of such a strategy.

1. Introduction

Water usage in biopharmaceutical processes is becoming an ever more present sustainability concern, as water used in the process needs to be of high grade. This means water is either purified water (PW) or water-for-injection (WFI) and producing PW or WFI is a water and energy intensive process [1–3]. Recycling of buffers used in biopharmaceutical production also across unit operations is therefore a concept that is under discussion, but current discussions around buffer recycling are unclear about the potential benefits. Estimates of 90 % are given, but detailed analysis is lacking and many concepts evolve around recycling buffers into the WFI production, which does not reduce the actual demand of WFI, or the associated energy [4,5]. A direct recycling of waste streams back into the production train across unit operations would save WFI directly and the associated cost in energy and water to produce it but has not been investigated on a merit-based risk aware basis.

Buffer recycling within one unit operation is already a concept accepted in biopharmaceuticals. This is true for those unit operations which inherently use buffer recycling in their operation, like simulated moving bed (SMB) [6–10]. Concerns of accumulation of byproducts like

host cell proteins (HCP) or host cell DNA within the process during operation have been periodically raised when buffer recycling is discussed, but are accepted by authorities. For instance, cyclic unit operations like periodic counter current chromatography (PCC) that are implemented on a large scale do have the potential of accumulation of byproduct or aberrant product forms [11–14]. In theory, antibody associated HCP or aberrant antibody forms could be handed from column to column indefinitely when using PCC where the flow-through of one column is connected to the next column in line [15]. For those unit operations where recycling or substantial back-mixing is already present and accepted, this acceptance is usually not based on merit, or risk awareness, but largely due to historic acceptance. It was accepted as-is without further discussion of the potential risk, because it was part of the concept of that unit operation and therefore unavoidable [16–18]. This acceptance could be extended to a risk-based assessment of buffer recycling across unit operations as well, but the potentially achievable savings are not quantified yet.

Recycling of buffers has also been discussed in terms of recycling buffer streams back into the preparation of new WFI [4]. While this can save some water in the WFI production, the large energy costs associated

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with WFI production is not addressed by this concept. This means that approaches of recycling waste streams into fresh WFI production do not have the potential for large savings in water usage, and have no potential in energy savings. In addition, they make WFI production significantly more complicated, as the inlet fluids might be already highly contaminated.

Direct buffer recycling back into the production across unit operations without going through WFI production, was viewed critically in the past, as the cost of water seemed negligible in comparison to the process costs. With the upcoming sustainability discussions across all sectors of industry, reducing WFI use in biopharmaceuticals is coming back into focus. While the idea itself might lead to assumptions of 50, 90 or even 100 % in water saving [5], the practical possibilities limited by risk awareness and technical feasibility in terms of compatible buffer types, are unknown. A direct recycling strategy across unit operations needs inlet streams that are noncritical in terms of process performance and product safety, as well as suitable outlet streams in other unit operations that are non-critical in terms of impurity load. In addition, both of them need to be compatible in terms of buffer design [4].

In this work, a strategy for risk-aware classification of inlet and outlet streams of a complete monoclonal antibody purification is presented and used to quantify the potential savings in WFI use.

2. Material and methods

2.1. Model system description tool

To define a standard monoclonal antibody production scheme a publicly available tool was used (mAbsizer™ calculator). No affiliation of the author exists to the company hosting the tool, but it was selected as the publication landscape does not frequently offer complete process descriptions. Such a description necessarily needs to include all process steps from harvest to final formulation including all materials and common safety factors in a publicly easily accessible form. As the monoclonal antibody process is consolidated in its form, the process scheme can be easily transferred to any other mAb and the described risk categorization can be expanded to any other bioprocess.

3. Results and discussion

3.1. Risk based buffer categorization

3.1.1. Characterization of inlet streams

For a rational approach on buffer recycling within the whole process chain and for categorizing the associated risk for buffer streams in a broad definition, three fundamental concepts are proposed.

First, only cleaning in place (CIP) solutions are available to be recycled into using forward recycling. This means that buffers can go from a unit operation into any further downstream unit operation only when being used for CIP. The rationale behind this approach is that CIP solutions are meant to contain impurities anyways during the process and are flushed out sufficiently to leave no significant residues. In addition, CIP solutions are perfectly capable to inactivate viruses, and therefore the risk for compromising viral safety is minimal.

The process steps are designed to handle impurities in a CIP solution, which means forward recycling is of minimal risk, when recycling into CIP steps. Forward cycling into any other inlet stream might pose the risk of impurities “skipping” one or more-unit operations and being reintroduced further downstream into operations that are not designed to handle them.

Second and following from that, any other recycled solutions may only be recycled backwards, meaning from a further downstream unit operation into a further upstream unit operation. The rationale being this is that any further downstream unit operation is cleaner and less burdened with impurities and any back cycled impurities will pass again any unit operation designed to remove them. Additionally, the

backcycling of buffers within one unit operation is an already broadly accepted concept.

And lastly, buffer recycling needs to take buffer compatibility into account, meaning that no complicated buffer adjustments like reverse osmosis or substantial dilution by fresh WFI are considered here. This means that designing a process with buffer compatibilities in mind, could mean higher potential in buffers savings.

After defining the general rationale behind the inter-unit-operation connections, individual buffer streams need to be further characterized according to their potential risk. There is a higher potential contamination risk if buffers are being used for column elution, than for re-equilibration, as the elution buffer will directly contain the product, and will be transported to the next unit operation.

In this work, buffers will be categorized into 4 different risk categories: A: High risk buffers that transport the product into the next unit operation; B: Medium risk buffers that are adjacent to category A and therefore have a reasonable chance of being intermixed with A and therefore also being transported into the next unit operation. C: Low risk buffers that fulfill a more technical role and are not next to category A buffers, and finally D: CIP solutions that are open for replacement by forward recycling as described above. In theory, only a small part of a buffer B is able to be transported further downstream, but for simplicity and technical feasibility, the whole buffer section will be considered category B with one notable exception: Flushing of filtration units will be split into two parts, whereas half will be considered A, and the rest will be considered C, meaning that the filters can be pre-flushed with recycled buffers to clear all air and remove impurities from the filter manufacturing, but will be flushed with WFI before use.

In this work only recycling into C and D category will be considered as a more conservative, risk-aware approach in light of a higher chance of implementation and future acceptance of the concept from regulatory perspective. Recycling into buffers that could be intermixed with the product might be seen as a critical risk and not be implemented at all.

3.1.2. Characterization of outlet streams

Some potentially replaceable buffers might lack any suitable process stream within the same process to be replaced by. To achieve replacement, it is necessary that both, a replaceable stream and a suitable stream for its replacement, are present in the process scheme.

Defining the buffers that can be recycled also needs a risk aware approach, as not all buffers are suitable for being recycled. For instance, strip buffers or CIP solutions should never be recycled, as they are intended to contain the most difficult to remove impurities in the respective process step and usually contain harsh buffer conditions. In this regard, the buffers can also be risk-classified. To avoid confusion with the A to D classification for inlet streams, outlet streams from a unit operation will be classified by R-S-T, with R being recyclable buffers of low risk (for instance equilibration buffers), S being buffers that have some risk associated as they contain impurities (like wash-buffers in chromatography) and T buffers which are of high risk like stripping buffers and CIP solutions. T and S buffers will not be considered in this work following a low-risk approach that could find acceptance by regulatory, and only R buffers are considered for recycling.

3.2. Model process

Bioprocesses can vary greatly even for similar products and therefore buffer consumptions may vary greatly. The current biggest productions are for monoclonal antibody, and these productions are both consolidated in their purification scheme and large enough to matter significantly in the bigger picture. For process definition a single-use based production train of 2000 liter fermentation is used with a titer of 5 g/L. Any additional use for CIP or sanitization in place (SIP) in a stainless-steel manufacturing setting would need to be addressed separately but is not the focus of this work.

The processes will be based on manufacturer recommendations for

how to treat filters and chromatography resins in the process, and a sample purification scheme is presented in Fig. 1. The typical mAb process consists of a depth and sterile filtration, followed by protein A chromatography, viral inactivation, two ion exchange chromatography steps, virus filtration and ultrafiltration diafiltration (UFDF) for formulation.

From a water perspective the virus inactivation does not have any significant impact, as it consists solely of the addition of acid/base or surfactant, which are in any case risk category A solutions and no waste stream is produced that could be recycled. Therefore, this step will be neglected in the analysis. For UFDF for formulation only flush water for the filter is considered, as any buffer used for UFDF is grade A as well, and the buffer composition can be highly complex which is problematic in the scheme of a direct recycling of the stream back into any other unit operation.

3.3. Case study mAb purification – mAbsizer™ model

To assess all potential feed streams, the individual waste streams and feed streams for each individual unit operation need to be identified and classified. The process used in this work is based on the “mAbsizer™ calculator” published by MerckMillipore (<https://www.merckmillipore.com/mabsizer-calculator>). The tool recommends specific column sizes, filtration areas, etc. but the specific recommended buffer compositions need to be taken from the respective manufacturer’s manual of the individual materials, or if not available from other similar materials. The process replaces the UFDF step with a simple UF, but otherwise adheres to the general scheme presented above (Fig. 2). From this scheme, the media preparation and bioreactor are ignored as well, as all buffers going into these steps are grade A and no significant waste stream is produced.

3.3.1. Harvest - depth and sterile filtration

Filtration for harvest clarification is a very common method, especially in single use setups. Large-scale centrifugation in single use is uncommon and has not been available in the past. The process definition calls for a 2-stage depth filtration with a total filter area of 40 m² which includes a safety factor, and an additional sterile filtration area of 1.8 m².

Filtration does not use any buffers during operation, but will require large amounts of water for flushing of the filters before use. Reports of used flushing volumes in production sites are rare, but a recent publication shows the use of at least 30 Liters of WFI for rinsing per m² of membrane in an optimized setup (fractionated flow-through for proofing the required < 500 µg/L total organic content). The same publication shows 300 Liters if the whole rinsing volume is collected and tested afterwards as a bulk [19,20]. The process definition for filtration already includes a recommended safety factor of two, taken from literature [19, 20]. For filter rinsing, WFI or phosphate buffered saline in WFI is recommended, but other buffers will work as well [19,20]. This rinsing waste stream can be problematic to recycle, as the rinse is used to reduce the total organic carbon content that is making its way into the product stream, and no upstream unit operation exists. This means that this process stream can only be forward cycled into CIP procedures. For a 2000 Liter process and the corresponding 41.8 m² of total filtration area,

the process uses an incoming stream of at least 1250 Liters in an optimized setup and 12,500 Liters in an unoptimized one. In this work, the assumption is that other water saving measures are already in place and an already optimized use of water of 1250 Liters for filter flushing is assumed. To address the potential risk associated with this process step, the stream will be split into two fractions (625 Liters each). Only the second half will be considered for forward recycling, as the total organic content of the first half of the flush is too high (Fig. 3) [19,20].

3.3.2. Capture chromatography

Capture chromatography is done according to the process scheme using Eshmuno® A resin with a binding capacity of 48 g/L and a recommended column size of 81 Liters with a total binding capacity of the column of 3.8 kg. It is unclear why the column is necessarily that large, as the total amount of antibody in the process is 10 kg, indicating that the column is only used for three cycles. Protein A materials have repeatedly been shown to withstand 100 or more cycles which makes this choice a very conservative one [21,22]. Nevertheless, the number of cycles is irrelevant in a purely water-centric analysis as long as the specific binding capacity of the column is the same. This means that low cycle numbers are not important in this study, but if any subsequent economic analysis is based on this analysis, this is an important point to consider. The recommended buffer scheme according to the manufacturer manual of the material is presented in Fig. 4. Wherever a range was given for buffer consumptions, it was assumed that an optimized process operates at the lowest given volumes. The figure also lists the risk categorization of each of the respective buffers and their total volumes in the process. The protein A chromatography capture uses a total of 1520 Liters of buffers, and 730 (48 %) of those are category C or D buffers, and therefore potentially replaceable by recycled solutions. Right before the protein A chromatography, the scheme recommends another 1.8 m² of sterile filtration that would use an additional 54 Liters of WFI. This step is presumably only necessary if the filtered harvest is stored before the capture and any particulates forming during storage need to be removed and is therefore omitted in this analysis.

3.3.3. Cation exchange chromatography

The mAbsizer™ tool offers three different resins (Eshmuno® CPX, Fractogel® COO and Fractogel® SO³⁻) as options, and Eshmuno® CPX is selected for this analysis as it offers the highest binding capacity and therefore the smallest column volume (47 Liter instead of 84). This will consequently use less buffers than the other suggested options. The choice was made in line with the idea that this analysis should present further improvements through recycling for an already water-consciously designed process.

Unfortunately, the user guides for Eshmuno® CPX only offer direct guidance on the elution buffers used (5–10 CV of pH 5.5, 50 mM sodium acetate with a gradient of sodium chloride from 40 to 1000 mM) but no direct guidance for equilibration, wash, and CIP. The material allows for an NaOH CIP, and this is the option that will be used here. With no volumes given, the respective volumes and schemes for the non-eluting phases are based on a similar cationic resin material from another manufacturer (Capto™ S ImpAct) as for that material the elution buffer is the same, and volumes and buffers are given for all other steps. In contrast to the recommended 5–10 CV linear elution for Eshmuno® CPX,

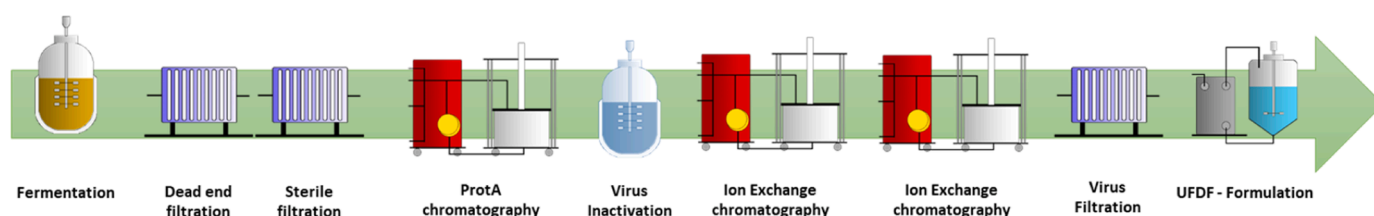


Fig. 1. Generalized process scheme of a monoclonal antibody production.

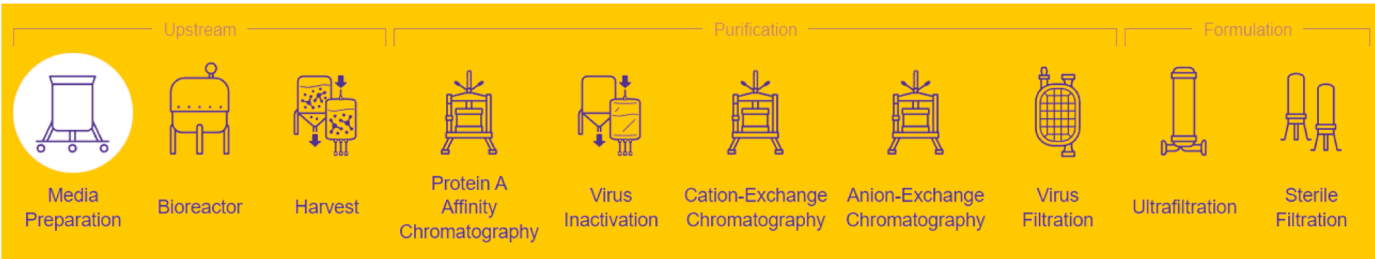


Fig. 2. Process scheme for the production of a monoclonal antibody according to the online tool mAbsizer™.

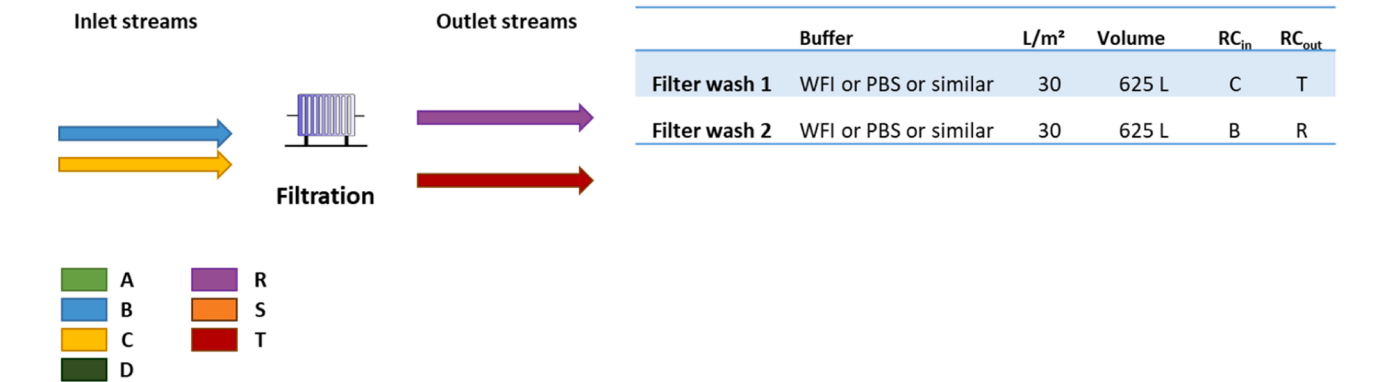


Fig. 3. Graphical representation of inlet and outlet streams as well as buffer volumes, use, composition and risk categorization (RC) for incoming streams (A, B, C, D) and outgoing streams (R, S, T) for depth filtration for cell removal.

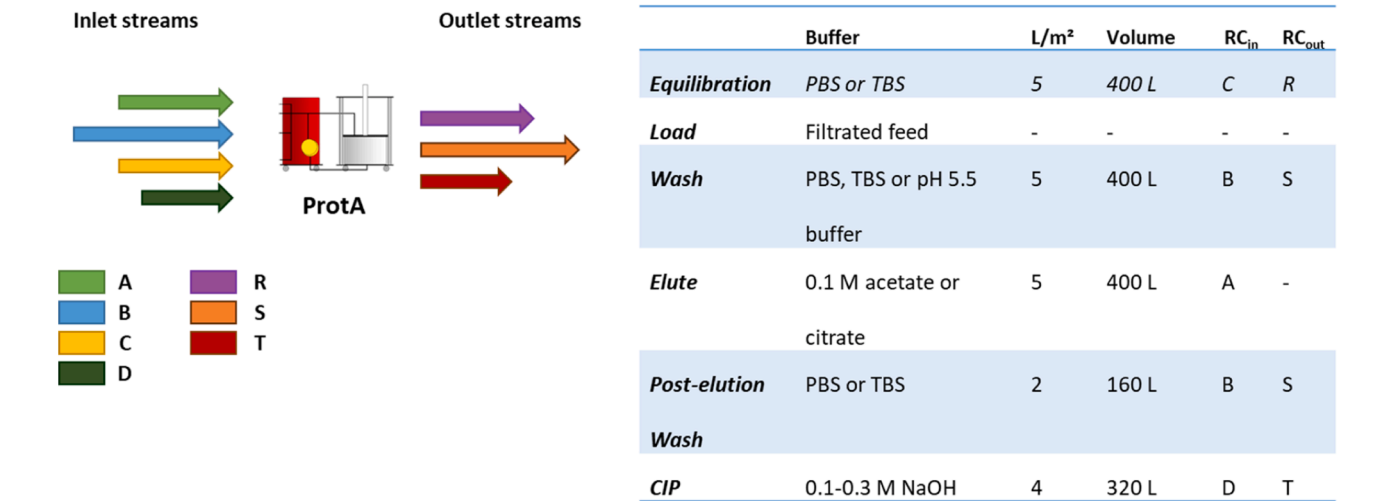


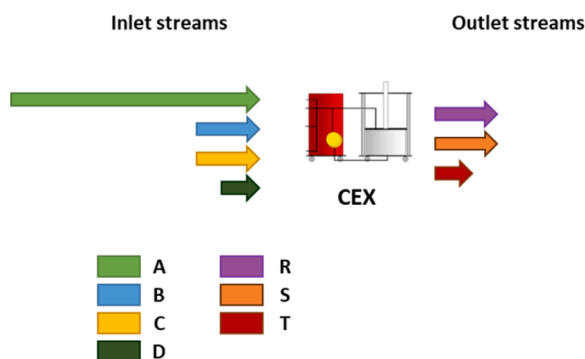
Fig. 4. Graphical representation of inlet and outlet streams as well as buffer volumes, use, composition and risk categorization (RC) for incoming streams (A, B, C, D) and outgoing streams (R, S, T) for protein A capture chromatography.

the manual of Capto™ S ImpAct mentions 20 CV and the mAbsizer™ tool assumes a total elution volume of 940 Liters (20 CV) as well. For this analysis, 20 CV of elution buffer is used, as this is what the tool assumes to be necessary and it was planned to stick to the tool as much as possible. For future process development, this may be the first significant savings that can be made in direct buffer consumption. The cation exchange bind-elute first polishing step uses a total of 1551 Liters of WFI, with 376 (24 %) of that being category C and D and therefore potentially replaceable (Fig. 5). The elution using 20 CV is already 60 % of the used buffers in this process step, and therefore has a potentially large optimization potential.

3.3.4. Anion exchange chromatography

Anion exchange chromatography needs a treatment of the eluate from the cation exchange chromatography to lower the conductivity to be able to load it successfully. In this case a 4 fold dilution is necessary according to the process definition, using a total of 2820 Liters of category A, as the solution is in direct contact with the product. This substantial WFI use can also be reduced significantly, if the linear gradient in the cation step is steeper and shortened to 10 CV or less.

Recommended materials for the cation exchange from the portfolio of the mAbsizer™ are Eshmuno® Q, Fractogel® TMAE and NatriFlo® HD-Q. NatriFlo® HD-Q is a membrane adsorber that needs a completely different volumetric setup (in CV buffers used for each step) than bead-



	Buffer	L/m ²	Volume	RC _{in}	RC _{out}
Equilibration	50 mM sodium acetate, 40 mM NaCl	5	235 L	C	R
Load	Filtrated feed	-	-	-	
Wash	50 mM sodium acetate, 40 mM NaCl	5	235 L	B	S
Elute	50 mM sodium acetate, 40-1000 mM NaCl	20	940 L	A	-
CIP	0.1-0.3 M NaOH	3	141 L	D	T

Fig. 5. Graphical representation of inlet and outlet streams as well as buffer volumes, use, composition and risk categorization (RC) for incoming streams (A, B, C, D) and outgoing streams (R, S, T) for cation exchange chromatography.

based methods, so in this case Eshumon® Q was selected as a standard material for polishing. It also assumes the same necessary column sizes as Fractogel® TMAE (43 Liters).

The recommended buffer system for equilibration of a flow-through anion exchange using this material is pH 7.5 according to the manufacturer using tris as a buffer system. There is no fundamental reason why this buffer could not be replaced by a phosphate-based buffer for equilibration. Use of a phosphate-based buffer would make re-use of the buffer stream easier as no other process step used Tris-buffered solutions. Again, no specific volumes are given for the respective buffers used, so this work will use the process scheme used by a competitor product (Capto™ Q) which offers a respective process scheme. The buffers used for wash and equilibration (25 mM phosphate) can reasonably be replaced by any pH 7–7.5 buffered solution without any expected large change to the process performance, as this step is a flow-through operation not binding the antibody. The process streams are shown in Fig. 6 and it uses a total of 946 Liters, with 645 in category C or D (68 %). In contrast to the first two bind-elute chromatographic steps, operating the AEX in flow-through mode to bind the impurities means that the equilibration buffer could be transported into the next unit operation by axial mixing, while the wash is unable to be transported into the next unit operation. Therefore, in flow-through chromatography, risk classes for equilibration and wash are reversed in comparison to bind-elute mode.

3.3.5. Virus filtration and ultrafiltration

Virus filtration is done following the final polishing step and is

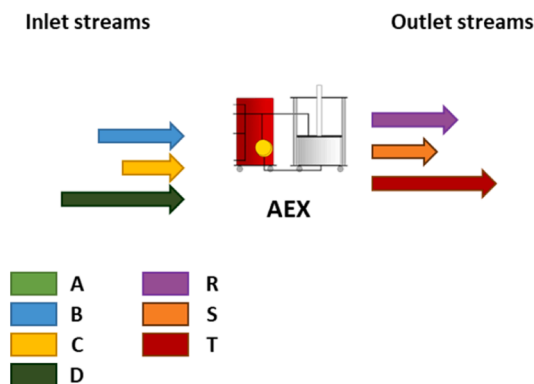
recommended with 2.1 m² of virus filtration membrane and 4.1 m² of sterile filter, resulting in a total rinse volume necessary for filter preparation of 186 Liters that can be potentially recycled into CIP fractions of preceding unit operations.

Additionally, the ultrafiltration uses a total of 27 m² of TFF membrane. Assuming the same rinse intensity as for depth filtration the TFF membrane flushing uses a necessary rinse volume of 810 Liters of WFI. It could be argued that the first half of the ultrafiltration membrane rinse could be treated analogous to the depth filtration rinse at the start of the process, but the risk of contamination in this step is significantly larger, and there is no process step downstream anyways, so the whole 810 Liters have to be fresh WFI. The ultrafiltration is again followed by a final sterile filtration of 1.4 m² using 42 Liters of WFI (Fig. 7).

3.3.6. Buffer usage process overview

With the assembled risk categorized buffers and defined process steps, an overview of the whole process can be assembled per unit operation and the respectively used buffers as well as the potentially recyclable buffers (Fig. 8). The respective buffer use is distributed throughout the unit operations, but the buffer risk categories are drastically different in the respective unit operations. Additionally, the recyclable buffers are distributed throughout the whole process chain, which is beneficial for a direct recycling concept, as available streams can more efficiently be recycled.

From the buffer composition that is needed (Fig. 9–1 and Fig. 9–3) and that is given in the process (Fig. 9–2 and Fig. 9–4), it can be seen that a total of 1970 Liters of buffers is recyclable (risk category R), but a total



	Buffer	L/m ²	Volume	RC _{in}	RC _{out}
Equilibration	25 mM phosphate	7	300 L	B	R
Load	Filtrated feed	-	-	-	
Wash	25 mM phosphate	5	215 L	C	S
Strip	25 mM phosphate, 1 M NaCl	5	215 L	D	T
CIP	0.1-0.3 M NaOH	5	215 L	D	T

Fig. 6. Graphical representation of inlet and outlet streams as well as buffer volumes, use, composition and risk categorization (RC) for incoming streams (A, B, C, D) and outgoing streams (R, S, T) for cation exchange chromatography.

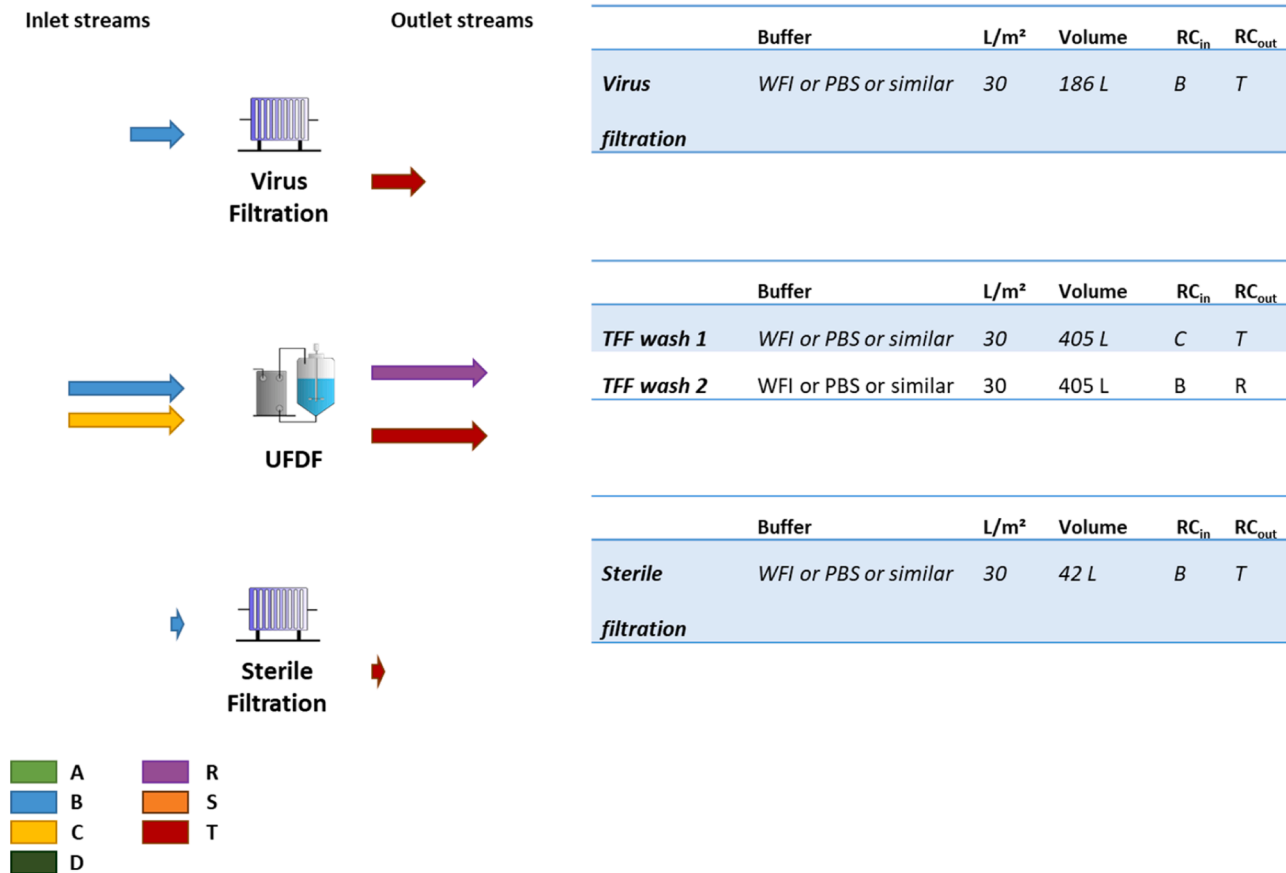


Fig. 7. Graphical representation of inlet and outlet streams as well as buffer volumes, use, composition and risk categorization (RC) for incoming streams (A, B, C, D) and outgoing streams (R, S, T) for virus filtration, ultrafiltration(TFF) and sterile filtration.

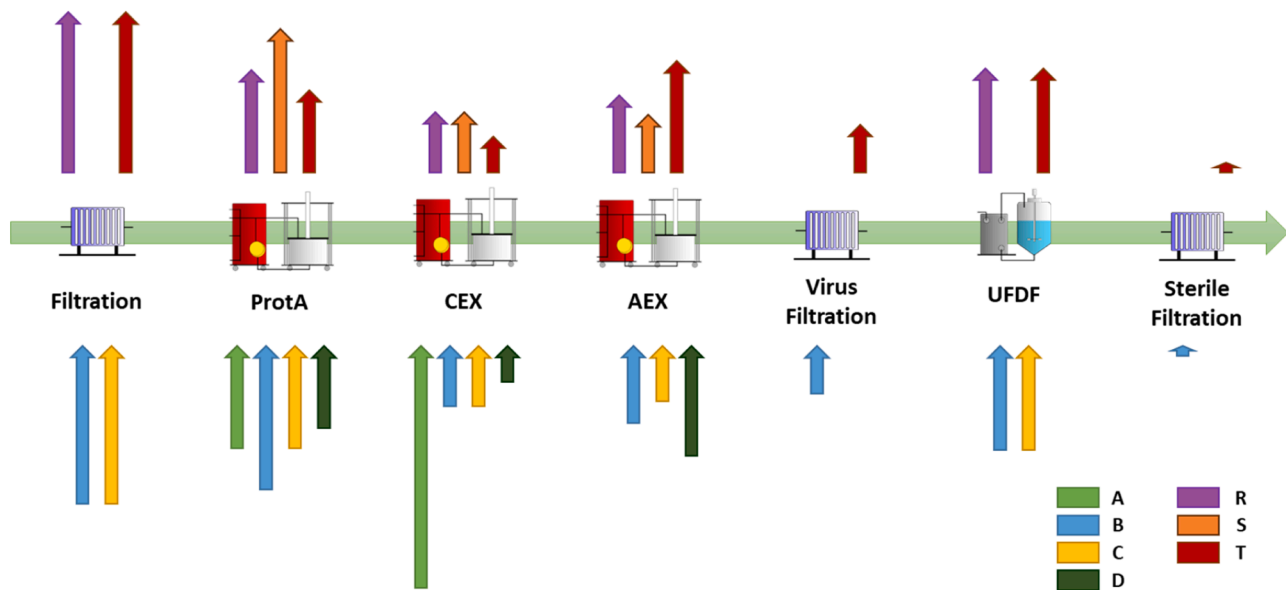


Fig. 8. Graphical representation of inlet and outlet and risk categorization (RC) for incoming streams (A, B, C, D) and outgoing streams (R, S, T) for the whole monoclonal antibody purification scheme.

of 2770 Liters could be accommodated throughout the process (risk category C and D). This means that there is too little reusable buffer that can be recycled, to fulfill the full potential of recycling in the process. From a total of 6460 liters used in the process scheme, 42 % of the buffers could be replaced potentially (2770 liters) but there are only 30

% of potentially available recyclable buffers in the process (1970 liters). This means process steps need to be prioritized for replacement of buffers. For this consideration category D buffers are replaced before replacing buffers in category C and replacing buffers in earlier unit operations is prioritized before replacing buffers in later unit operations.

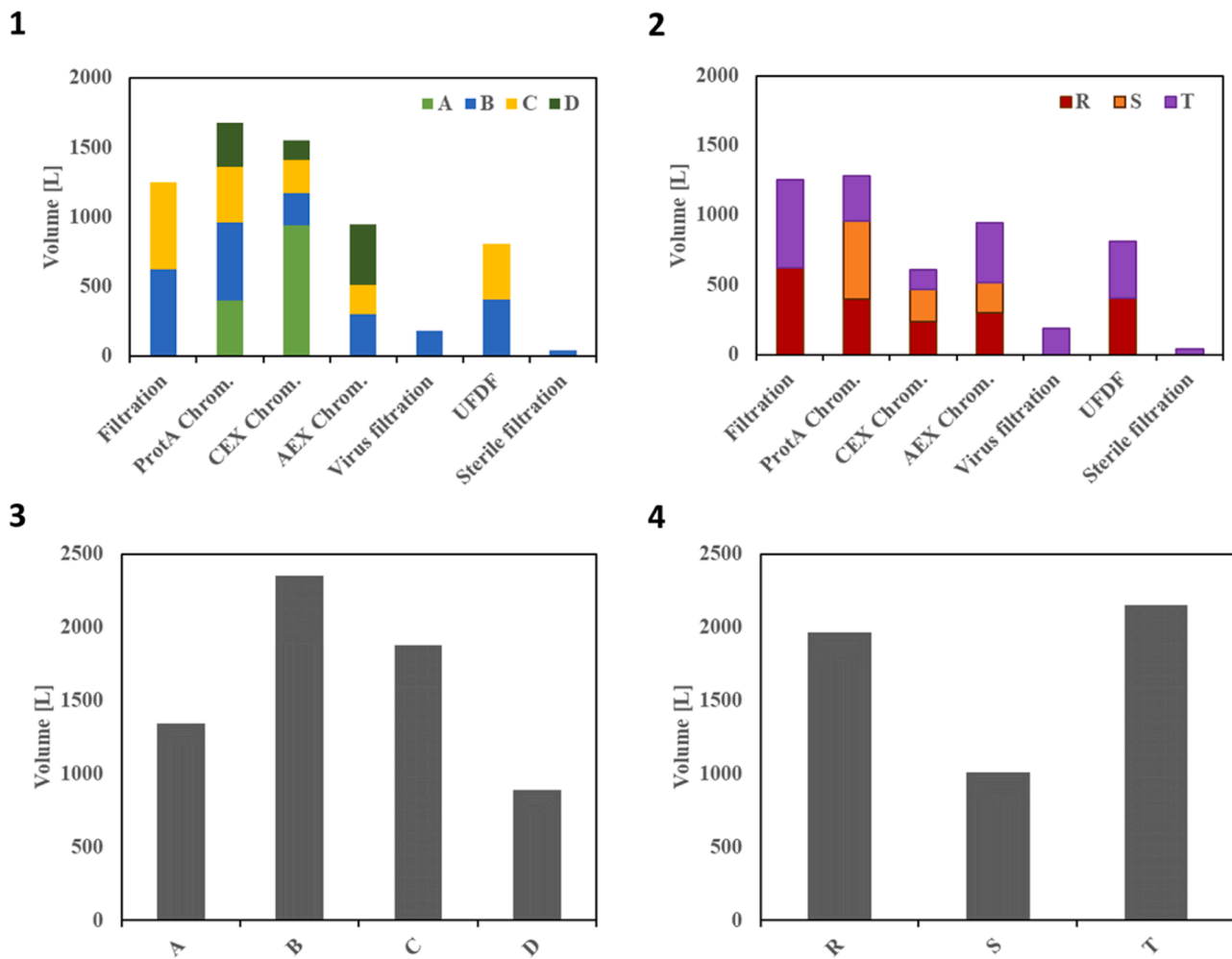


Fig. 9. Buffer volumes of different risk categories throughout the process for the respective inlet streams (1) and outlet streams (2) and total amount of buffers used in all steps for inlet streams (3) and outlet streams (4) split into their respective risk categorization (RC) for incoming streams (A, B, C, D) and outgoing streams (R, S, T).

3.3.7. Final replacement strategy

According to the prioritization, the first thing to replace are the CIP solutions and they can also be replaced by forward recycling. The process produces a total of slightly over 1000 Liter of flush water from filtrations throughout the process that is category R and uses a total of almost 900 Liters of category D solution. As category D solutions are allowed to be forward cycled into as well, there is no restriction in reusing buffers from the first depth filtration for that purpose. Allowing for preparation of excess CIP solution, this already uses up all respective R category buffers from the depth filtration by replacement of all CIP solutions which saves a total of 13.7 % of WFI.

With priority given to earlier process steps, the depth filtration in the process uses a total of 625 Liters of potentially replaceable category C buffer. Which can be replaced by 400 Liters of phosphate buffered equilibration buffer from the protein A, and an additional 300 Liters of phosphate buffered equilibration buffer from the anion exchange chromatography. In theory, the 235 Liters of acetate buffered equilibration from the cation exchange chromatography could be used as well, but mixing different buffer solutions could have unexpected effects and phosphate buffered solutions are non-critical for flushing of depth filtrations. Allowing for the preparation of excess buffer, the phosphate buffered equilibration solutions of the process are already used up with this single recycling loop that allows for an additional 9.7 % reduction of WFI use.

This only leaves the acetate buffered equilibration used in the cation

exchange chromatography, but with 235 Liters this is not enough to replace the category C phosphate wash buffer pH 5.5 in the upstream protein A chromatography (needed 400 Liters) and this would necessitate a respective change in the unit operation and development of the method to accommodate an acetate equilibration. If this step of additional process development would be taken, this would result in an additional saving of 3.6 %, which might not be worth it.

Future changes in process design for recycling more water within the process could include a harmonization of all involved buffers across unit operations instead of developing each unit operation individually. As shown, this approach is very limited in what it can achieve, as only the last recycling loop saving 4 % of WFI was restricted by the buffers used. The approach of buffer harmonization might still make buffer preparations simpler in the process, but the WFI savings in enables through recycling are limited.

Additionally, outgoing streams could be split into multiple streams of different risk categories whenever warranted, as this investigation showed that we lack suitable recyclable streams to be used. The process is not lacking in terms of streams to be recycled into and this approach could yield an additional 10 % savings.

An additional stream not represented in this analysis is the excess buffer preparation during manufacturing as usually more buffer is prepared, than is actually needed in the process. This excess preparation is hard to quantify, as companies do not share this information readily, but leftover buffer from one unit operation can be forward cycled without

adding risk to the process, and can be used to top up the preparation of other buffers. Recycling across batches, or even recycling across different products could increase the potential water savings dramatically, but especially cross-product recycling will be under strict scrutiny by regulatory due to safety concerns.

In practice, this means that with a low-risk approach to recycling buffers directly in the process, only two recycling loops can be formulated, one for replacing CIP solutions with water used for flushing filters, and a second one recycling equilibration buffers from the chromatography units into the flush of the depth filter (Fig. 10). While the first one is perfectly feasible for any setup, the second one can be complicated in practical terms. Chromatography units allows for buffer collections after each cycle, making buffer transport from subsequent units at least possible in theory if all chromatography unit operations are run at the same time in cycles. This is not possible for depth filtration, as it is done in one cycle. This means that any buffers that needs to be recycled into the depth filtration from subsequent unit operations need to be from a previous purification run. This is only possible in a campaign setting of multiple fermenters being harvested in one day, or within a couple of days, where buffers are stored in between.

As an alternative to cycling into the depth filtration, one can imagine a recycling into the C-buffer of the protein A purification, and while it might be technically challenging with current equipment and setups, it is at least possible to do so within one purification run. This alternative would save 4.5 % instead of 9.7 % and is not be worth the technical difficulties.

Depending on how cross-batch buffer recycling is seen, the risk-based assessment for recycling shows that 14 % of WFI savings can be realized quite easily, 18 % could be achieved with some technical difficulties, and 23 % can only be achieved if cross-batch recycling is allowed. An additional 4 % savings could be achieved when unit operations are designed for better buffer compatibility between unit operations but yet again only with some technical difficulties.

This risk aware assessment of the potential of recycling buffers across

unit operations to save WFI shows that the benefits that can be realized are potentially significantly lower than the expected benefits. With a risk sensitive approach, no further WFI savings beyond 27 % can be realized, even when cross-batch recycling is allowed, technical difficulties overcome, and processes are redesigned for buffer compatibility. This is because there are not enough streams in the process that could be recycled, even though there are some streams that could still be replaced. This shows that practical risk associated limitations for buffer recycling are substantial and reduce the actual savings of buffers from estimates of 90 % in the past [5] down to maybe as low as 14 % if risk to the product and concerns about batch cross-contamination are considered.

This highlights indirectly the importance of continuous manufacturing, and the more promising approach of buffer recycling within one unit operation. While impurity accumulation through buffer recycling has to be characterized more stringently in the future, the potential savings can be substantially larger. Unfortunately, such concepts are still unable to save the WFI used in filter flushes, which make up 35 % of the total process scheme (2290 Liters out of 6500 Liters). This limits WFI savings by buffer recycling within unit operations to 65 %, even if 100 % buffer recycling would be achievable for all chromatography steps. If CIP solutions cannot be recycled, the potentially achievable recycling drops to 50 % even if all other buffers are perfectly recycled.

The results of this work are not specific on any specific mAb, but if the general process scheme depicted here is changed, the potential savings in the process can change significantly. The process scheme for mAb is significantly consolidated in this platform process. Alternative approaches like flow-through chromatography [23,24] as well as alternatives to protein A chromatography are under development [25–34]. While these technologies can change the outcome of this analysis, they are not yet widely implemented in industry. Nevertheless, the methodology presented here, the risk classification of inlet and outlet streams and the constraints formulated for forward and

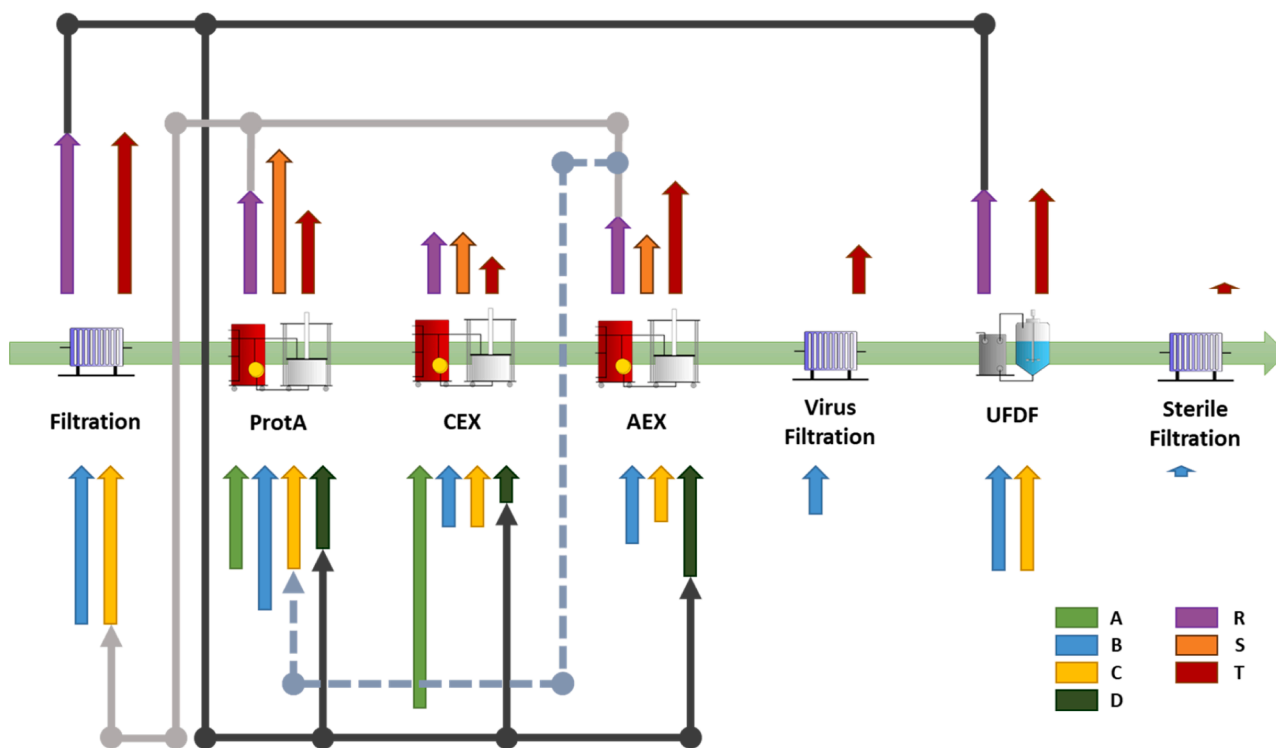


Fig. 10. Possible recycling loops using risk-aware design for potential direct recycling throughout the process split into their respective risk categorization (RC) for incoming streams (A, B, C, D) and outgoing streams (R, S, T). Black recycling line represents the recycling into strip buffers, gray the recycling of wash buffers, and the dotted blue line the recycling of the rest of the AEX equilibration buffer into the protein A.

back-cycling are applicable for other processes beyond mAb manufacturing.

In conclusion, the main goal has to be the reduction of WFI use within the process, followed by reuse of buffers into the flushing of filters, and lastly, the recycling within unit operations for the rest of the buffers. Neglecting cross-unit operation recycling of buffers limits the potential maximum to 50 % of recycling, while a combination with recycling across unit operations can unlock additional potentials. Still, 90 % or more recycling efficiency will be hardly achievable, as impurities need to exit the process at some point, contained within a respective buffer stream like the CIP solutions.

4. Conclusion

In this work a risk-based approach to categorizing all inlet and outlet buffer streams from all unit operations in a monoclonal antibody purification scheme showed the maximum potential for WFI savings is substantially limited. While 43 % of the buffers used in the process scheme were identified to be replaceable with minimal risk, only a fraction of that can actually be realized. Past estimated or hopes of savings of 70, 80 or even 90 % were too optimistic and did not take risk to the product or practical considerations, like CIP solutions leaving the process into account. While 43 % still shows a huge potential, a detailed analysis for the available waste streams available for replacing these buffers showed that only 23 % savings could be realized when risks are considered and cross-batch recycling is allowed. Savings are dropping further down to 14 % if cross-batch recycling is not allowed. The analysis also showed that flush volumes for filters are very significant with 35 % of the total volumes. While recycling within unit operations was demonstrated in the past with high efficiencies, these large volumes and the fact that CIP solutions need to leave the process, limit these approaches to 50 % of the total buffer volumes of the process, even if all non-CIP solutions are perfectly recycled. The findings of this study highlight the importance of combining all three strategies to achieve high WFI savings: Reduction of buffers used in the first place, recycling across unit operations for filter flushes, and buffer recycling within other unit operations like chromatography. Only a combined strategy can hope to achieve WFI savings above 50 %. The risk categorization used in this work to assess ingoing and outgoing streams is universal and the methodology can be used for other production schemes. It can therefore serve as a standard to quantify potential WFI savings using buffer recycling across unit operations in biologics manufacturing.

CRedit authorship contribution statement

The manuscript has only one author and all contributions made to the article were only done by myself. This includes according to CRedit: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, and Writing – review & editing. Funding acquisition is not applicable in this case, as no external funding was received. Software is not applicable as no specialized software was developed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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